

BUILDING CRACK DETECTION USING IMAGE RESTORATION AND SEGMENTATION TECHNIQUES

Submitted by:

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1. PROBLEM STATEMENT

Cracks in concrete buildings and structures can indicate deterioration and the need for inspection and maintenance. Identifying cracks manually from concrete surface images can be time-consuming and may become difficult when images contain noise, surface texture, shadows, or other unwanted variations.

Digital Image Processing (DIP) techniques can be used to improve the quality of images and identify crack-like regions. This project develops a simple image-processing pipeline for building crack detection using image restoration and segmentation techniques.

Seven concrete crack images are selected from a publicly available dataset. The images are converted to grayscale and controlled salt-and-pepper noise is introduced to evaluate restoration techniques. Mean, Median, and Gaussian filters are then applied and compared using Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM).

The best-performing restoration output is subsequently used for crack segmentation using Otsu thresholding and morphological operations. Canny Edge Detection and Adaptive Thresholding are also implemented as additional techniques for comparison.

2. OBJECTIVES

The main objectives of the project are:

1. To study the application of Digital Image Processing for building crack detection.
2. To use concrete crack images as input data.
3. To convert RGB images into grayscale images.
4. To introduce controlled salt-and-pepper noise into the images.
5. To apply Mean, Median, and Gaussian filters for image restoration.
6. To evaluate the restoration techniques using MSE, PSNR, and SSIM.
7. To compare the restoration results using graphs and visual outputs.

8. To select a suitable restored image for segmentation.
9. To segment crack-like regions using Otsu thresholding.
10. To improve the segmentation output using morphological operations.
11. To analyse the final segmentation results.
12. To study Canny Edge Detection and Adaptive Thresholding as additional segmentation approaches.
13. To compare the outputs of different segmentation techniques.
14. To understand the limitations and possible improvements of DIP-based crack detection.

3. DATASET / IMAGE DESCRIPTION

3.1 Dataset

The project uses the Concrete Crack Images for Classification dataset available through Mendeley Data.

Dataset: Concrete Crack Images for Classification

Version: 2

Source: Mendeley Data

DOI: 10.17632/5y9wdsg2zt.2

The complete dataset contains 40,000 concrete surface images, including:

- 20,000 positive crack images
- 20,000 negative images

The images were collected from concrete structures at METU Campus Buildings.

The dataset images have a resolution of 227×227 pixels and are RGB colour images.

3.2 Images Selected

For this project, 7 positive crack images were selected from the dataset.

The selected images were renamed as:

- crack_01.jpg
- crack_02.jpg
- crack_03.jpg
- crack_04.jpg
- crack_05.jpg
- crack_06.jpg
- crack_07.jpg

3.3 Preprocessing

The following preprocessing operations were performed:

1. The RGB images were loaded using OpenCV.
2. The images were converted into grayscale.
3. Approximately 5% salt-and-pepper noise was artificially added.
4. Mean, Median, and Gaussian filters were applied.
5. Restoration performance was evaluated using MSE, PSNR, and SSIM.
6. The median-restored images were used for the main segmentation pipeline.

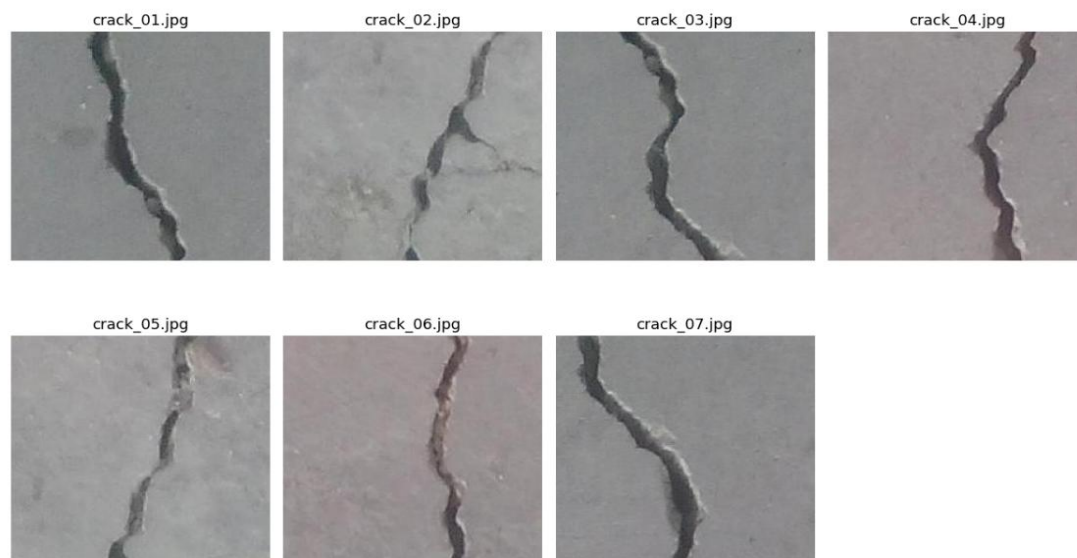


Figure 1: Original Concrete Crack Images

4. METHODOLOGY

The overall methodology followed in the project is:

Input Images → Grayscale Conversion → Salt-and-Pepper Noise Addition → Image Restoration → MSE/PSNR/SSIM Evaluation → Otsu Thresholding → Morphological Processing → Final Crack Segmentation → Additional Segmentation Techniques → Comparison

4.1 Original Crack Images

Seven positive concrete crack images were selected from the dataset and used as the input images.

The original images contain concrete surface regions with visible cracks.

4.2 Grayscale Conversion

The RGB images were converted into grayscale images using OpenCV.

Grayscale conversion reduces the image to a single intensity channel and simplifies the subsequent filtering and segmentation operations.

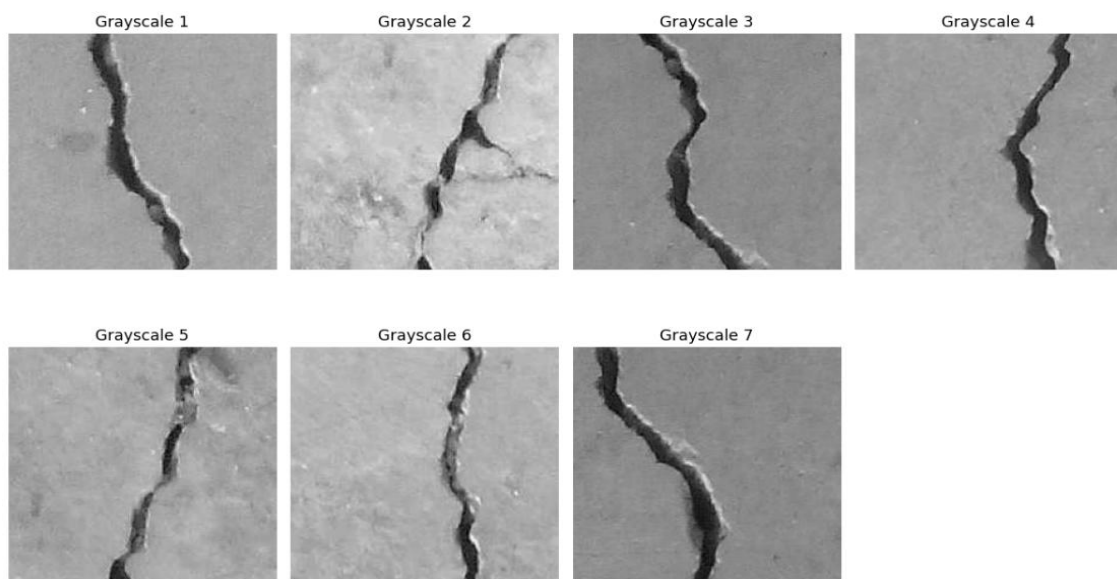


Figure 2: Grayscale Images

4.3 Salt-and-Pepper Noise Addition

Salt-and-pepper noise was artificially added to the grayscale images at approximately 5%.

This noise introduces random black and white pixels into the image. The purpose of adding controlled noise was to test the ability of different restoration filters to recover the original image.

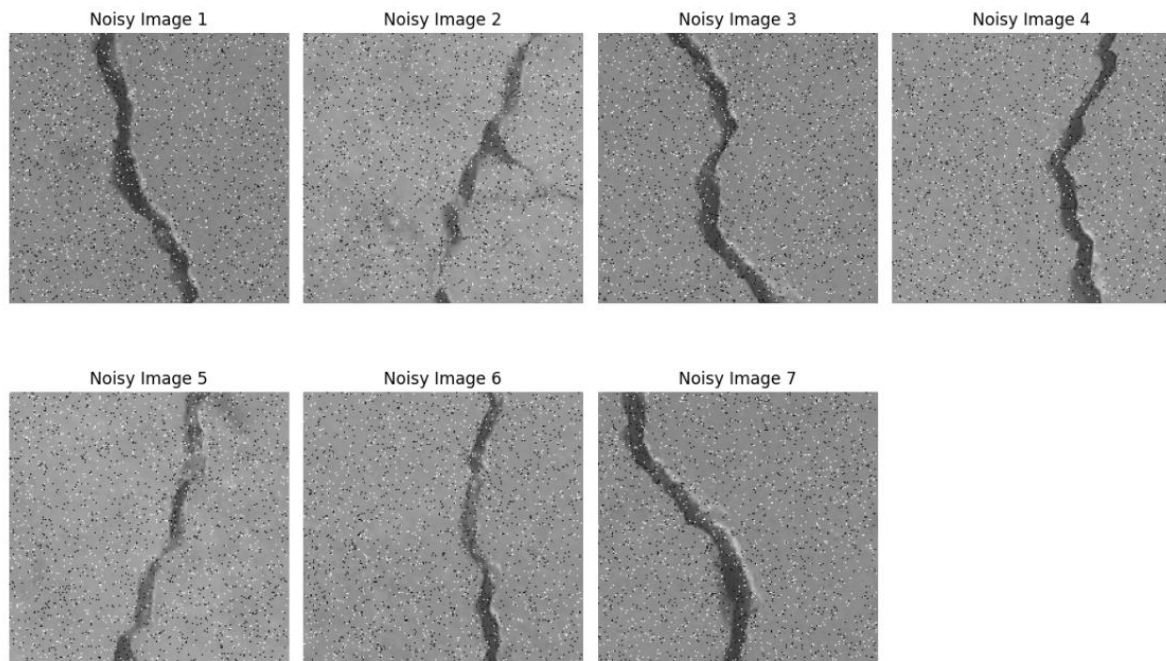


Figure 3: Grayscale Images with Salt-and-Pepper Noise

5. IMAGE RESTORATION

Three spatial filtering techniques were used for restoration.

5.1 Mean Filtering

A 5×5 Mean Filter was applied to the noisy images.

The mean filter replaces each pixel with the average of the neighbouring pixel values. It can reduce noise but may also smooth edges and fine details.

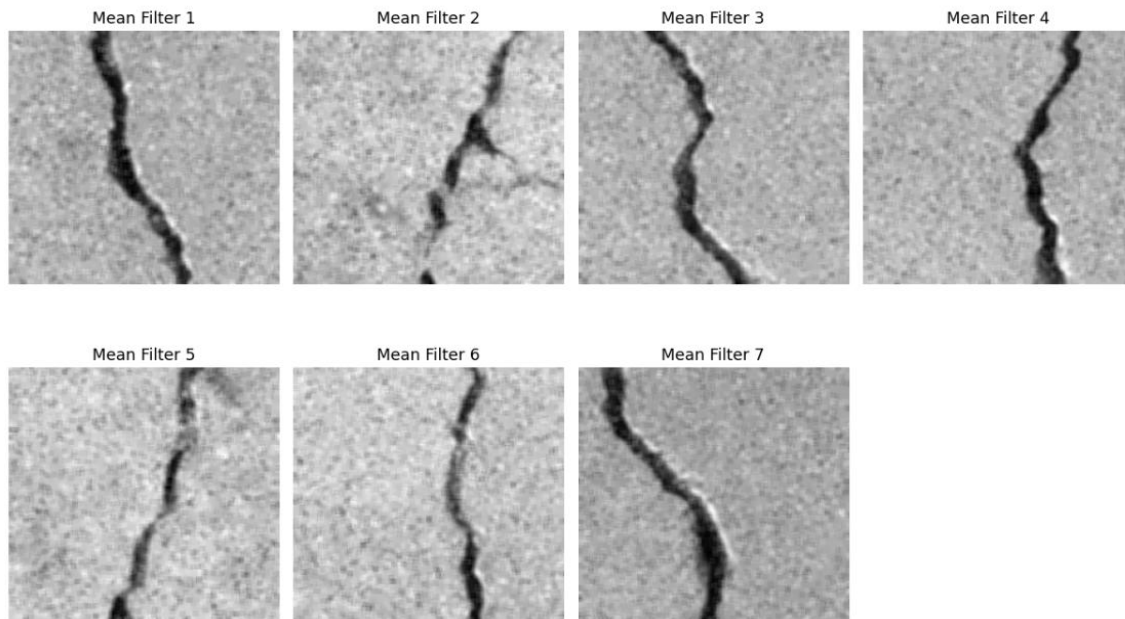


Figure 4: Images after Mean Filtering

5.2 Median Filtering

A 5×5 Median Filter was applied to the noisy images.

The median filter replaces each pixel with the median value of its neighbourhood. This type of filter is particularly suitable for impulse or salt-and-pepper noise.

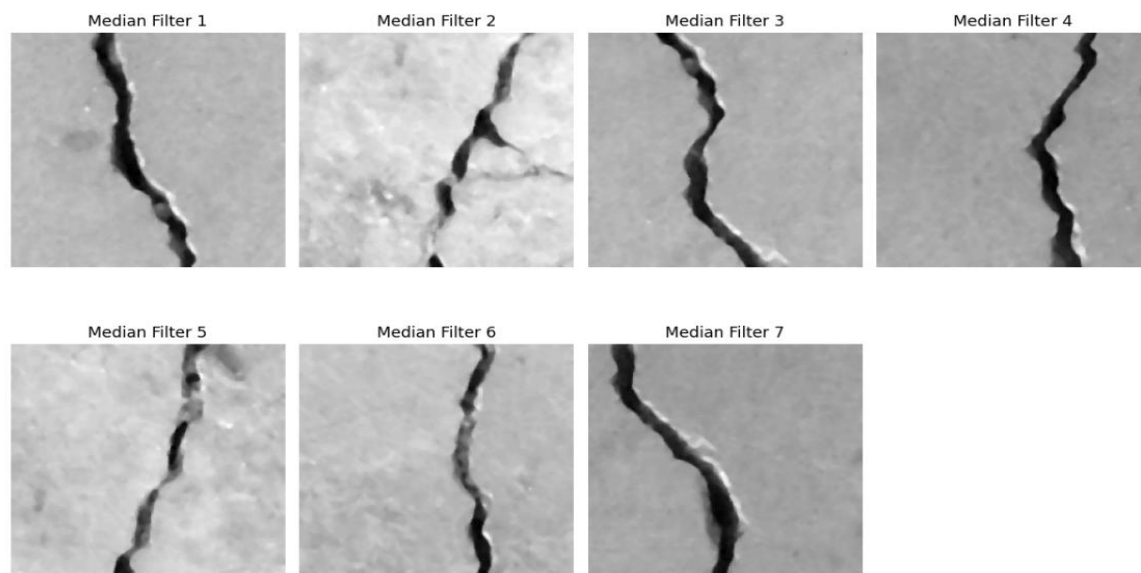


Figure 5: Images after Median Filtering

5.3 Gaussian Filtering

A 5×5 Gaussian Filter was applied to the noisy images.

Gaussian filtering performs weighted smoothing, giving more importance to pixels closer to the centre of the neighbourhood.

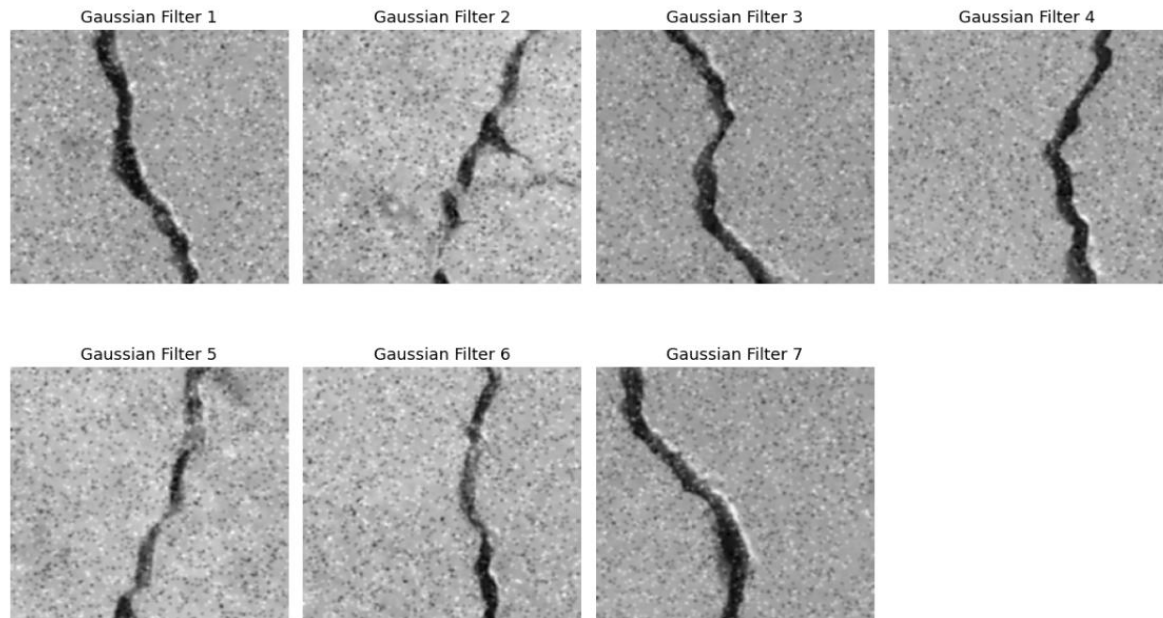


Figure 6: Images after Gaussian Filtering

6. RESTORATION COMPARISON

The outputs from the Mean, Median, and Gaussian filters were visually compared.

The comparison helps to observe how each filter removes the artificially introduced noise while retaining the important structures in the crack images.

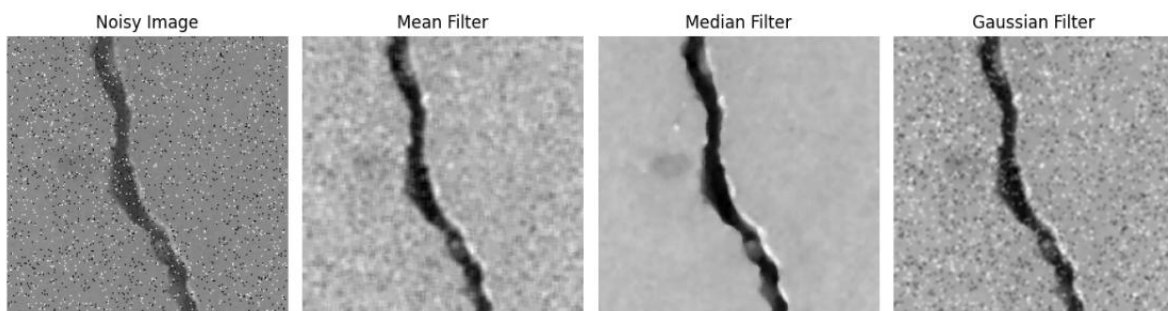


Figure 7: Comparison of Restoration Techniques

7. RESTORATION EVALUATION

The restoration techniques were quantitatively evaluated using:

Mean Squared Error (MSE)

MSE measures the average squared difference between the original and restored images.

A lower MSE indicates a smaller difference between the original and restored images.

Peak Signal-to-Noise Ratio (PSNR)

PSNR measures the quality of the restored image relative to the original image.

A higher PSNR generally indicates better restoration quality.

Structural Similarity Index (SSIM)

SSIM measures the structural similarity between two images.

A value closer to 1 indicates greater structural similarity.

7.1 Restoration Results

The average results obtained from the seven images are:

<u>Filter</u>	<u>Average MSE</u>	<u>Average PSNR (dB)</u>	<u>Average SSIM</u>
Mean	50.13	31.13	0.7171
Median	9.97	38.16	0.9392
Gaussian	70.71	29.64	0.6180

The Median Filter produced the lowest MSE and the highest PSNR and SSIM among the three techniques for the seven images used in this experiment.

Therefore, the median-restored images were selected for the main segmentation stage.

7.2 MSE, PSNR and SSIM Results

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Restoration Evaluation Results
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Image 1
Mean Filter      : MSE = 49.95  PSNR = 31.15  SSIM = 0.7103
Median Filter    : MSE = 10.67  PSNR = 37.85  SSIM = 0.9265
Gaussian Filter  : MSE = 70.67  PSNR = 29.64  SSIM = 0.6146

Image 2
Mean Filter      : MSE = 50.51  PSNR = 31.1   SSIM = 0.7168
Median Filter    : MSE = 10.03  PSNR = 38.12  SSIM = 0.9347
Gaussian Filter  : MSE = 72.35  PSNR = 29.54  SSIM = 0.6219

Image 3
Mean Filter      : MSE = 50.48  PSNR = 31.1   SSIM = 0.7193
Median Filter    : MSE = 10.87  PSNR = 37.77  SSIM = 0.9373
Gaussian Filter  : MSE = 69.41  PSNR = 29.72  SSIM = 0.6216

Image 4
Mean Filter      : MSE = 49.66  PSNR = 31.17  SSIM = 0.7253
Median Filter    : MSE = 9.21   PSNR = 38.49  SSIM = 0.9501
Gaussian Filter  : MSE = 68.26  PSNR = 29.79  SSIM = 0.6202

Image 5
Mean Filter      : MSE = 51.23  PSNR = 31.04  SSIM = 0.7145
Median Filter    : MSE = 9.88   PSNR = 38.18  SSIM = 0.9379
Gaussian Filter  : MSE = 73.44  PSNR = 29.47  SSIM = 0.6158

Image 6
Mean Filter      : MSE = 48.7   PSNR = 31.26  SSIM = 0.7261
Median Filter    : MSE = 8.56   PSNR = 38.81  SSIM = 0.9524
Gaussian Filter  : MSE = 69.37  PSNR = 29.72  SSIM = 0.6211

Image 7
Mean Filter      : MSE = 50.07  PSNR = 31.13  SSIM = 0.7214
Median Filter    : MSE = 10.92  PSNR = 37.75  SSIM = 0.9344
Gaussian Filter  : MSE = 69.32  PSNR = 29.72  SSIM = 0.6256
```

Figure 8: MSE, PSNR and SSIM Results

7.3 Average Restoration Results

The average values were calculated across all seven images.

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Average Restoration Performance
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Mean Filter
Average MSE   : 50.09
Average PSNR  : 31.13
Average SSIM  : 0.7191

Median Filter
Average MSE   : 10.02
Average PSNR  : 38.14
Average SSIM  : 0.939

Gaussian Filter
Average MSE   : 70.4
Average PSNR  : 29.66
Average SSIM  : 0.6201
```

Figure 9: Average Restoration Results

8. SEGMENTATION USING OTSU THRESHOLDING

After restoration, Otsu's thresholding method was applied to the median-filtered images.

Otsu's method automatically determines a threshold value from the intensity distribution of an image.

Inverse binary thresholding was used so that detected crack-like regions appear as white regions against a black background.

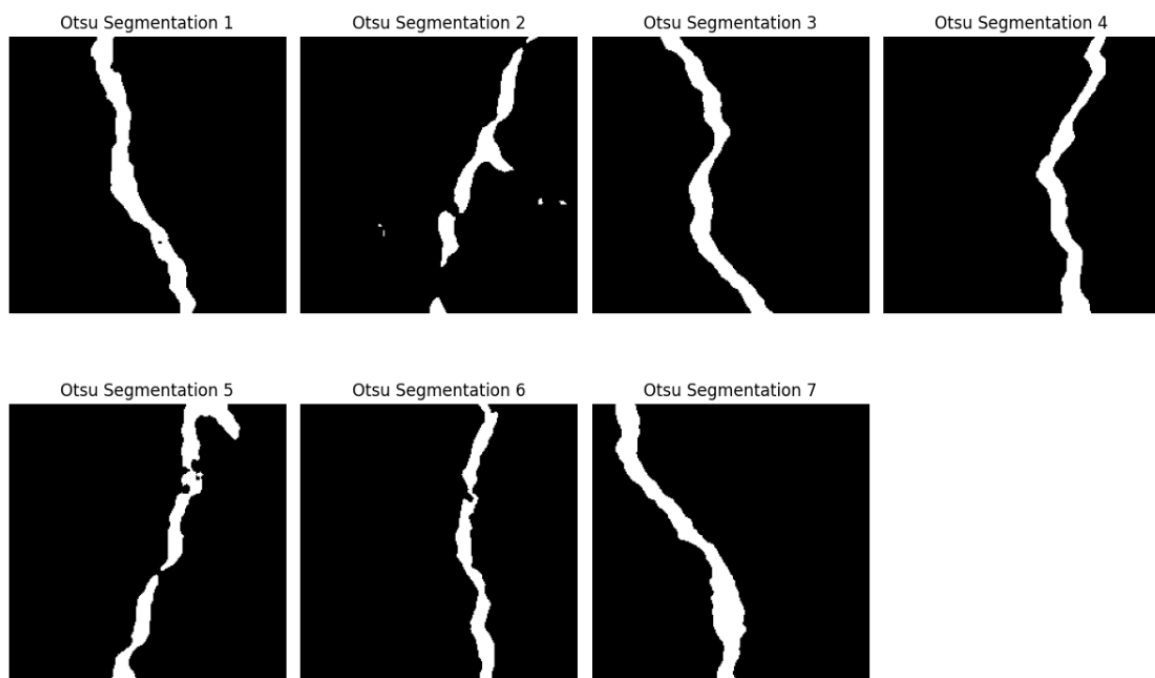


Figure 10: Crack Segmentation using Otsu Thresholding

9. MORPHOLOGICAL OPERATIONS

Morphological operations were applied to improve the Otsu segmentation.

A 3×3 structuring element was used.

Two operations were performed:

Opening

Opening was applied to remove small unwanted regions and noise.

Closing

Closing was applied to fill small gaps and connect nearby regions.

The combination of these operations produced a cleaner binary segmentation result.

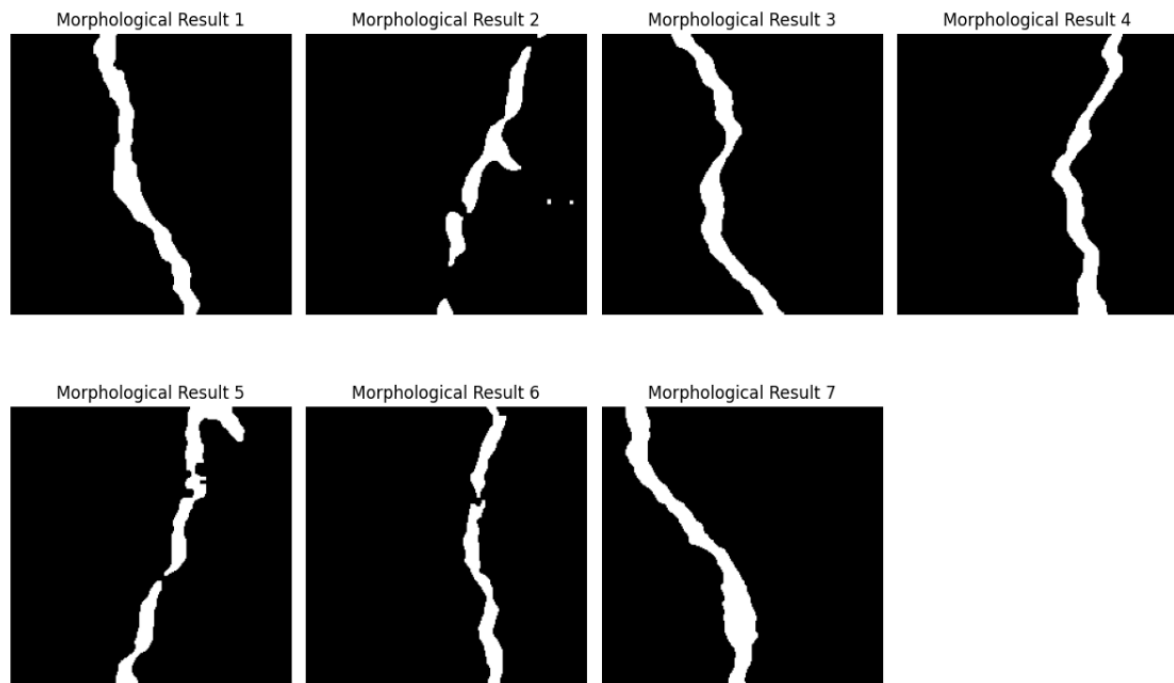


Figure 11: Results after Morphological Operations

10. BEFORE AND AFTER COMPARISON

The complete processing pipeline can be visualized by comparing the original image, noisy image, restored image, and final segmentation.

This comparison shows how the image changes at different stages of the proposed method.

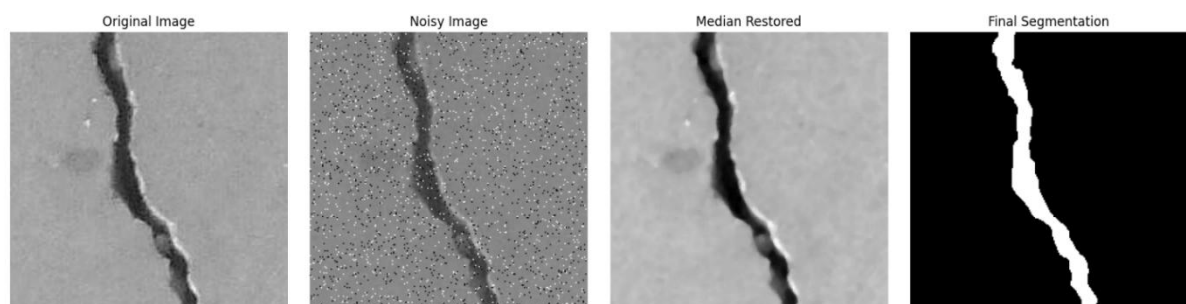


Figure 12: Before and After Comparison of the Crack Detection Process

11. FINAL CRACK SEGMENTATION

The final crack segmentation was obtained by applying Otsu thresholding followed by morphological opening and closing.

The resulting binary images represent the detected crack-like regions.

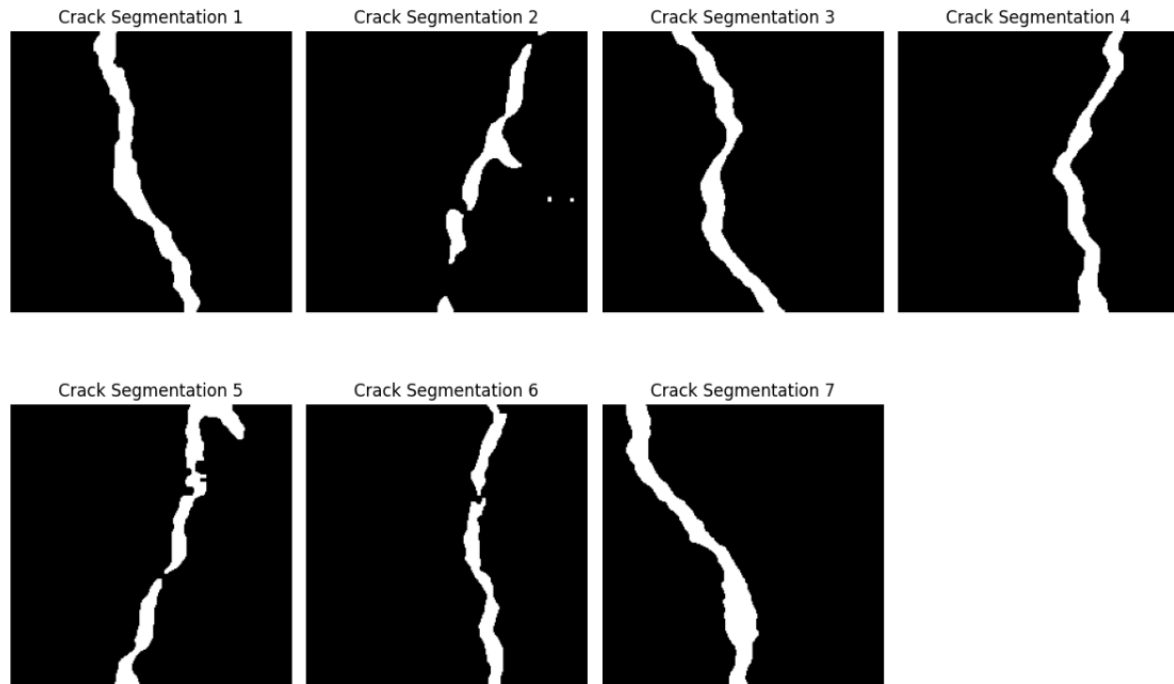


Figure 13: Final Crack Segmentation

12. RESTORATION PERFORMANCE COMPARISON

The restoration methods were compared using graphical representations of their average MSE, PSNR, and SSIM values.

The graphs provide a visual comparison of the performance of Mean, Median, and Gaussian filters.

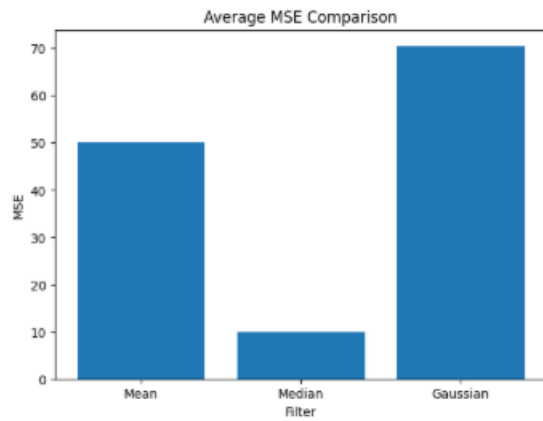


Figure 14: Average MSE Comparison

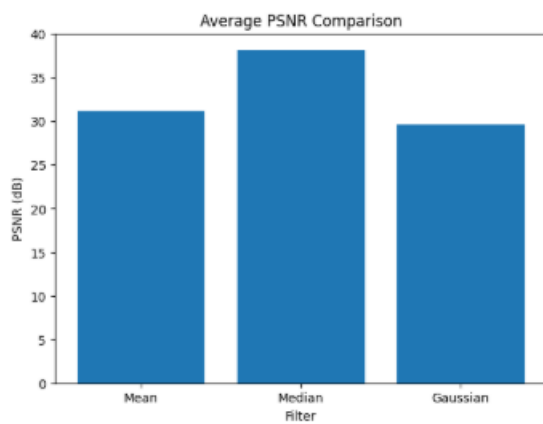


Figure 15: Average PSNR Comparison

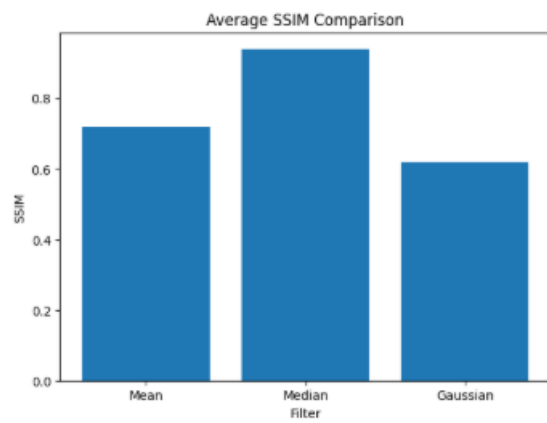


Figure 16: Average SSIM Comparison

13. SEGMENTATION ANALYSIS

The final segmented images were analysed by calculating the number of white pixels in each binary image.

The percentage was calculated as:

$$\text{Detected Region Percentage} = (\text{Number of White Pixels} / \text{Total Number of Pixels}) \times 100$$

This value indicates the proportion of the image classified as the detected region.

It should not be interpreted as the exact physical percentage of crack area because the segmentation may also include shadows, stains, surface texture, and other regions with similar intensity.

Since manually labelled ground-truth masks were not available, segmentation accuracy measures such as IoU and Dice coefficient were not calculated.

```
Crack Segmentation Analysis
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Image 1 : Detected Crack Region = 6.61 %
Image 2 : Detected Crack Region = 4.46 %
Image 3 : Detected Crack Region = 6.41 %
Image 4 : Detected Crack Region = 5.91 %
Image 5 : Detected Crack Region = 5.76 %
Image 6 : Detected Crack Region = 4.47 %
Image 7 : Detected Crack Region = 7.49 %
```

Figure 17: Segmentation Analysis of the Seven Images

14. SEGMENTATION COMPARISON

The median-restored image, Otsu segmentation, and morphological result were compared to observe the effect of the post-processing stage.

Morphological processing helps to remove some small unwanted regions and improve the binary segmentation.

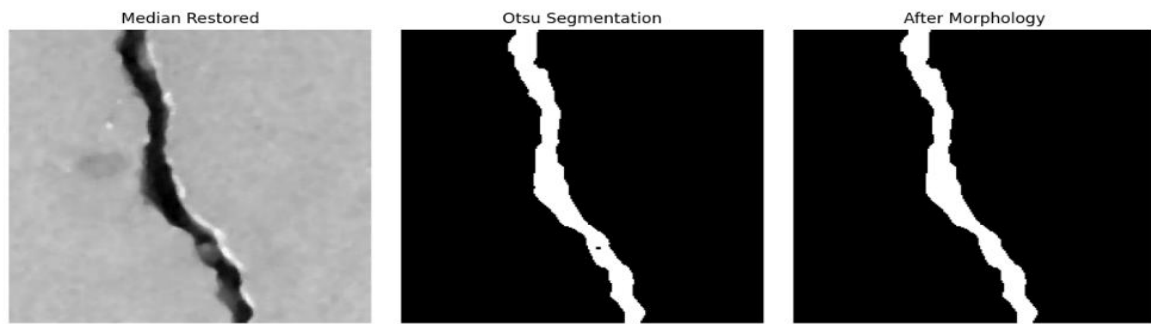


Figure 18: Comparison of Median Restored Image, Otsu Segmentation and Morphological Result

15. CANNY EDGE DETECTION

Canny Edge Detection was implemented as an additional image segmentation-related technique. Canny detects significant intensity changes in an image and represents them as edges.

The following threshold values were used:

- Lower threshold = 50
- Upper threshold = 150

The resulting edge images were visually analysed to observe how crack-like structures are represented.

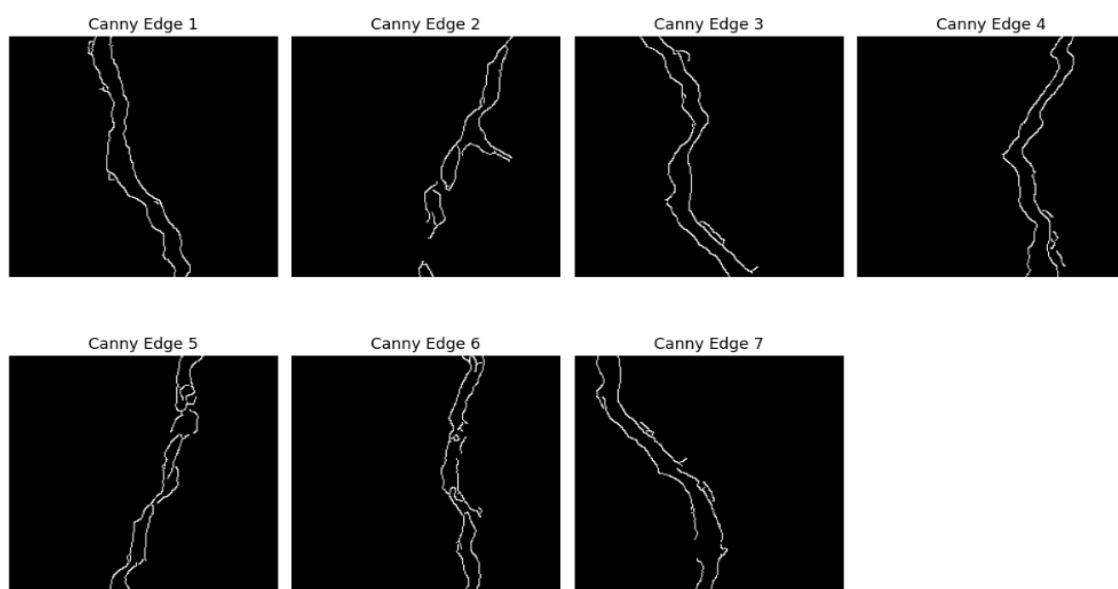


Figure 19: Crack Detection using Canny Edge Detection

16. ADAPTIVE THRESHOLDING

Adaptive Thresholding was also implemented as an additional segmentation technique.

Unlike global thresholding, adaptive thresholding calculates threshold values based on local regions of the image.

Gaussian adaptive thresholding was used with:

- Block size = 11
- Constant = 2

This method allows different areas of the image to use different threshold values.

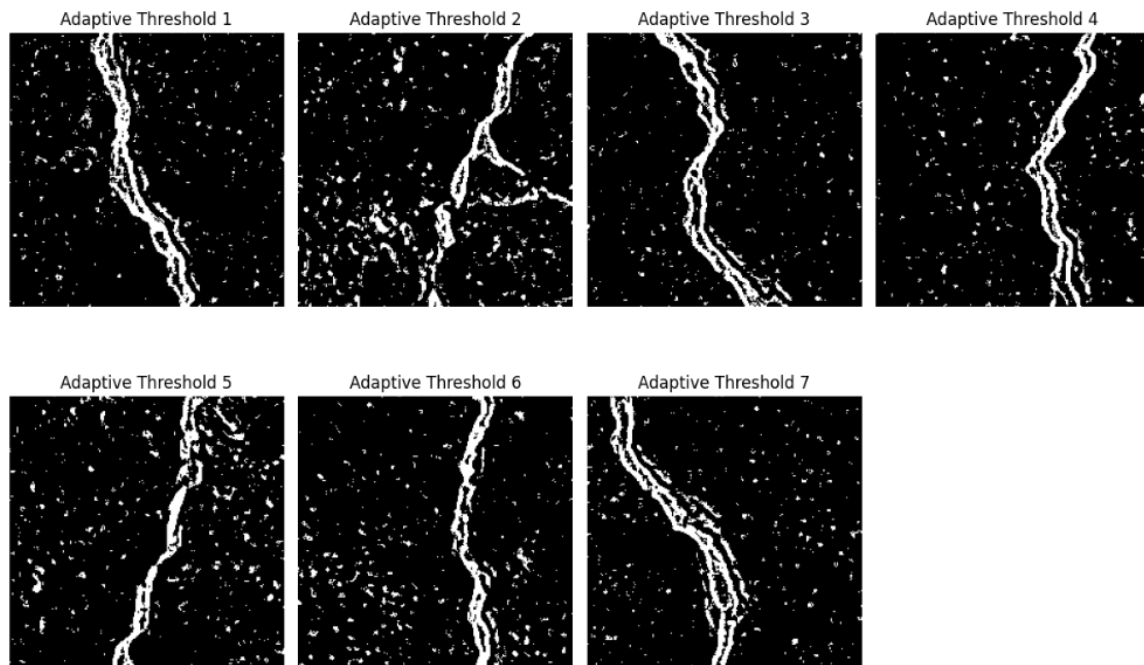


Figure 20: Crack Segmentation using Adaptive Thresholding

17. SEGMENTATION TECHNIQUE COMPARISON

The three segmentation approaches were compared:

Otsu Thresholding

Otsu thresholding produces a binary segmentation based on a global automatically calculated threshold.

Canny Edge Detection

Canny identifies edges and intensity boundaries in the image. It provides edge information rather than a filled crack-region mask.

Adaptive Thresholding

Adaptive thresholding calculates thresholds locally and can respond to variations in image intensity.

The comparison helps to understand how different segmentation methods behave on concrete crack images.

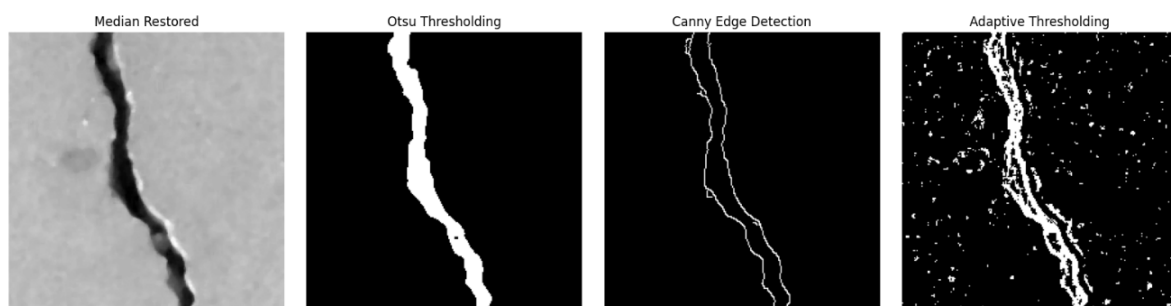


Figure 21: Comparison of Otsu, Canny and Adaptive Thresholding

18. OVERALL RESULTS AND DISCUSSION

The project successfully implemented a complete Digital Image Processing pipeline for building crack detection.

The restoration stage compared Mean, Median, and Gaussian filtering under controlled salt-and-pepper noise.

The obtained average results were:

Restoration Technique Comparison

Filter	MSE	PSNR (dB)	SSIM
Mean	50.13	31.13	0.7171
Median	9.97	38.16	0.9392
Gaussian	70.71	29.64	0.6180

For the seven selected images and the artificial salt-and-pepper noise used in this experiment, the Median Filter produced the strongest quantitative restoration results.

The restored images were then segmented using Otsu thresholding. Morphological opening and closing were applied to improve the binary output.

Canny Edge Detection and Adaptive Thresholding were also studied as additional segmentation approaches. Their outputs were compared visually with Otsu thresholding.

19. LIMITATIONS

The project has the following limitations:

1. Only 7 positive crack images were used in the experiment.
2. Salt-and-pepper noise was artificially introduced rather than using naturally noisy images.
3. The restoration experiment mainly focused on salt-and-pepper noise.
4. Otsu thresholding may detect shadows, stains, surface texture, or other regions in addition to cracks.
5. Ground-truth crack masks were not available for the selected images.
6. Therefore, segmentation metrics such as IoU and Dice coefficient could not be calculated.
7. Fixed filter sizes and threshold parameters were used.
8. Different concrete surfaces and lighting conditions may produce different results.
9. The selected images have a resolution of 227×227 pixels, which may limit the representation of very fine crack details.

20. FUTURE SCOPE

The project can be further improved by:

1. Using a larger and more diverse dataset.
2. Using real-world noisy concrete images.
3. Testing additional noise models and restoration techniques.
4. Using adaptive filtering techniques.
5. Experimenting with different segmentation methods.
6. Using ground-truth masks to calculate IoU, Dice coefficient, and accuracy.
7. Exploring advanced edge and region-based segmentation techniques.
8. Applying machine learning and deep learning-based crack segmentation.
9. Developing a real-time crack detection system.
10. Developing a web or mobile application for crack detection.
11. Testing the system under different lighting and environmental conditions.
12. Comparing traditional DIP methods with deep learning approaches.

21. CONCLUSION

This project presented a Digital Image Processing approach for building crack detection using image restoration and segmentation techniques.

Seven positive concrete crack images were selected from the Concrete Crack Images for Classification dataset. The images were converted from RGB to grayscale and approximately 5% salt-and-pepper noise was introduced for restoration testing.

Mean, Median, and Gaussian filters were applied to the noisy images. The restoration techniques were evaluated using MSE, PSNR, and SSIM. The Median Filter obtained an average MSE of 9.97, PSNR of 38.16 dB, and SSIM of 0.9392 for the seven images used in the experiment.

The median-restored images were then processed using Otsu thresholding and morphological opening and closing operations to obtain crack-like segmentation regions.

Canny Edge Detection and Adaptive Thresholding were additionally implemented and compared with Otsu thresholding.

The project provided practical experience in image restoration, spatial filtering, image-quality evaluation, threshold-based segmentation, morphological processing, edge detection, and comparative analysis. It also demonstrated the importance of selecting image-processing techniques according to the characteristics of the input images and noise.

22. GITHUB REPOSITORY

The complete project source code, input images, screenshots, and required files are available in the GitHub repository.

<https://github.com/sivanijbiju/DIP-Building-Crack-Detection.git>